



Incorporating modality-specific intensity prior as text prompt for multimodal myocardial pathology segmentation

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ABSTRACT

Accurate myocardial pathology segmentation from multimodal CMR is crucial for myocardial infarction risk assessment and treatment planning. Unlike traditional methods that rely on morphological and geometrical prior for assistance, we propose a novel intensity-guided multimodal image segmentation (I-MMSeg) method for myocardial pathology segmentation in this paper. Specifically, we leverage multimodal large language models (e.g., GPT-4o) to generate modality-specific intensity prior. Such prior capture both the relative intensity orders across different regions and the intrinsic intensity characteristic around the boundary of each pathology, serving as high-level guidance for the segmentation process. To this end, we adopt a CLIP-based model to encode the intensity prior of different modalities, and design two dedicated modules to incorporate them into the segmentation architecture: the intensity-prior-guided cross-modal feature enhancement module and class feature modulation module. The former promotes adaptive recalibration of visual feature and cross-modal feature interaction, while the latter modulates the textual class feature of each modality under the guidance of the intensity prior. Together, they enhance the discriminability of visual feature. Extensive experiments demonstrate that our proposed I-MMSeg method outperforms state-of-the-art myocardial pathology segmentation approaches. The source code and the self-collected MyoPS380 dataset are available at: <https://github.com/zzzzzzl24/I-MMSeg>.

1. Introduction

Myocardial infarction is a major public health issue that contributes to high morbidity and mortality rates. Accurate myocardial pathology segmentation is essential for risk assessment and personalized therapy (Thygesen et al., 2007; Reed et al., 2017). Cardiac Magnetic Resonance (CMR) imaging, as a non-invasive imaging technique with high tissue resolution, provides multimodal structural and functional information. Specifically, the balanced steady-state free precession (bSSFP) modality clearly delineates myocardial structural boundaries. The late gadolinium enhancement (LGE) modality highlights scar regions with signal intensity significantly higher than normal myocardium due to the retention of gadolinium-based contrast agents (Cui et al., 2024). The T2-weighted (T2w) modality emphasizes edema regions, which appear brighter than normal myocardium and exhibit diffuse and transitional characteristics. These complementary modalities offer rich cues

for lesion identification (Ding et al., 2023). However, due to differences in imaging mechanisms, different CMR modalities exhibit significant cross-modal heterogeneity in intensity distribution. Moreover, pathological regions such as scar and edema present distinct signal characteristics across modalities. As shown in Fig. 1, these differences are visually evident. Such modality-specific intensity variations hinder the learning of unified feature representations and the effective interaction of cross-modal features. Therefore, efficiently extracting and integrating key features from multimodal CMR to achieve accurate myocardial pathology segmentation remains a fundamental challenge.

Recent research on myocardial pathology segmentation has progressively transitioned from traditional image processing methods (Amado et al., 2004; Alba et al., 2012) to deep learning-based models (Li et al., 2023b; Fahmy et al., 2020). To enhance multimodal interaction,

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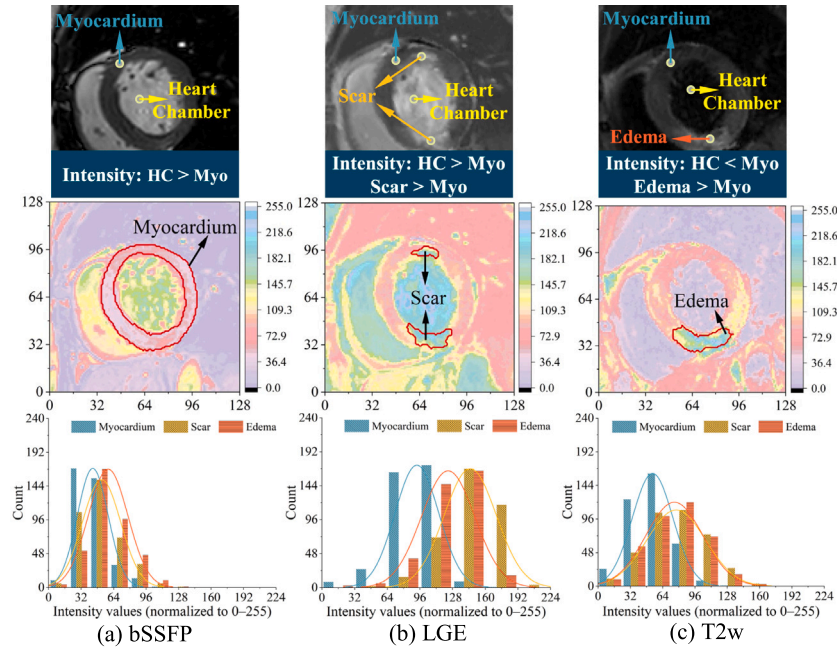


Fig. 1. Multimodal CMR intensity characterization of myocardium, scar, and edema. Top row: original image; Middle row: pixel heat map; Bottom row: intensity distribution of each class, computed from categorical mean values per sample.

various strategies have been proposed, including cascading architectures (Thaler et al., 2024), confidence gating (Liu et al., 2025), and Bayesian fusion (Xu et al., 2025), aiming to improve the utilization of complementary information across modalities. Meanwhile, several studies have incorporated morphological and geometrical prior to enforce anatomical consistency. For instance, Zhu et al. (2023) integrated a 3D V-Net with a shape deformation module for left ventricular segmentation. Qiu et al. (2023) employed an inclusive loss to model the spatial relationship between scar and edema. Existing morphological prior typically rely on stable anatomical structures and high-quality cross-modality registration, which limits their capacity to model fine-grained pathologies such as scar and edema. In contrast, intensity prior derived from inter-region differences in multimodal images are intuitive, robust, and generalizable. However, such intensity prior have received limited attention in the literature. Besides, how to effectively integrate these general and interpretable intensity prior into a unified multimodal image segmentation architecture remains an open problem.

The emergence of multimodal large language models (MLLMs) (H. Liu et al., 2023; Zhao et al., 2023; Zhang et al., 2023; Xiang et al., 2025; Luo et al., 2025) presents a transformative opportunity to bridge this clinical-technical gap. These models can generate human-understandable textual descriptions of modality-specific intensity relationships (e.g., “Compared with the myocardium, scar appears significantly brighter in LGE”) that encapsulate essential diagnostic knowledge. By leveraging the text encoding capabilities of BiomedCLIP (Zhang et al., 2023), modality-specific intensity prior can be systematically integrated into the visual feature learning process. Building on this foundation, we propose a novel intensity-prior-guided multimodal image segmentation (I-MMSeg) method with two key modules: (1) **Intensity-Prior-Guided Cross-Modal Feature Enhancement**: This module integrates modality-specific intensity prior with semantic information of visual feature to dynamically generate spatial weighting matrix, which adaptively calibrates the original visual feature and enhances the perception of modality-relevant regions. Then, the cross-modal information interaction is performed on the calibrated visual feature. (2) **Intensity-Prior-Guided Textual Class Feature Modulation**: This module integrates intensity prior and visual feature to generate linear modulation parameters for modulating modality-specific textual class feature.

The main contribution of this paper are:

- (1) We novelly model the intensity prior on relative order across different regions and the intrinsic intensity characteristic around the boundary of each pathology as explicit prior knowledge and incorporate it into the segmentation architecture, effectively improving myocardial pathology segmentation.
- (2) We propose the intensity-prior-guided feature enhancement and textual class feature modulation modules to strengthen visual and textual feature representation.
- (3) Extensive validation on multimodal CMR datasets demonstrates significant performance gains over some state-of-the-art methods. In addition, we also develop a challenging MyoPS380 dataset for myocardial pathology segmentation.

2. Related work

2.1. Myocardial pathology segmentation

Myocardial pathology segmentation aims to automatically identify pathological tissues such as scar and edema in myocardial regions, which is important for clinical diagnosis and treatment (Li et al., 2023a). Recent advances have used deep learning and multimodal CMR imaging, integrating bSSFP, LGE, and T2w modalities to improve segmentation accuracy. Innovative models such as MyoPS-Net (Qiu et al., 2023), which employs cross-modal feature fusion, and contrast TRustworthy prototype learning (CTRL) architecture (Liu et al., 2025), have been developed. To mitigate inter-modality misalignment, MvMM (Zhuang, 2018) proposes a probabilistic architecture that jointly models segmentation and registration. The U-MyoPS (Ding et al., 2023) design a combined computing model for simultaneous image registration and information fusion. AWSnet (Wang et al., 2022) introduces reinforcement learning for auto-weighted supervision, refining segmentation granularity. The MyoPS Challenge organized at the MICCAI conference (Li et al., 2023b) has further advanced this research direction by attracting several representative strategies tailored to realistic clinical scenarios. These include a coarse-to-fine paradigm that first identifies the foreground before segmenting pathological regions (Zhai et al., 2020), a stacked BCDU-Net with data augmentation fusion (Martín-Isla et al., 2020), deep convolutional models equipped with weighted bi-directional feature pyramids (J. Zhang et al., 2020),

data augmentation schemes that construct pseudo-color channels by combining multiple modalities (Ankenbrand et al., 2020), and cascaded frameworks that integrate anatomical and pathological segmentation (Z. Zhang et al., 2020). These approaches have played an important role in driving progress in multi-modality myocardial pathology segmentation. However, significant intensity differences across modalities, the small proportion of lesion regions, and the limited dataset size still severely constrain segmentation performance. To address these challenges, we propose I-MMSeg that integrates modality-specific intensity prior into segmentation and self-collected MyoPS380 dataset to tackle this problem.

2.2. Multimodal medical image segmentation

Multimodal medical image segmentation is an important research direction. Existing fusion strategies are generally classified into early, middle, and late fusion. Early fusion methods, such as Fuse4Seg (Guo and Su, 2024) and AdaFuse (Gu et al., 2023), integrate multimodal information at the input level through direct concatenation or attention in the frequency domain. Middle fusion approaches focus on feature-level integration. In particular, Yang et al. (2023) propose a unified hyper-network architecture that simultaneously performs multi-modal image synthesis and tumor segmentation, enhancing both structural coherence and segmentation accuracy. Zhou et al. (2022) design a network that incorporates spatial, channel, and modality-specific attention mechanisms to facilitate collaborative representation and fusion of cross-modal features. Transformer-based models like mmFormer (Zhang et al., 2022), M²FTTrans (Shi et al., 2023), and IMS²Trans (Zhang et al., 2024) further capture global dependencies to ensure structural and semantic consistency. Al Khalil et al. (2023) effectively address heterogeneity and outlier cases in cardiac magnetic resonance images through heart region detection and late fusion strategies. Despite great progress, fully leveraging multimodal complementarities in complex tasks like myocardial pathology segmentation remains challenging. Our I-MMSeg introduces modality-specific intensity prior to guide the fusion and segmentation process, improving the accuracy of myocardial pathology segmentation.

2.3. Integrating prior knowledge for image segmentation

The introduction of the prior knowledge can effectively improve the performance and interpretability of the segmentation model. Traditional prior mainly focus on shape prior (Zotti et al., 2018; J. Liu et al., 2023), anatomical structure constraints (Simantiris and Tziritis, 2020), and spatial consistency modeling (Meng et al., 2024), with anatomical prompt (Huang et al., 2024), topological prior (Gupta et al., 2022, 2024), mathematical representations (e.g., signed distance maps Li et al., 2020) to prompt and encode. In multimodal settings, conventional prior-based methods often rely on rigid definitions and require precise cross-modal registration, limiting their applicability under complex pathological conditions. Recently, multimodal large language models (MLLMs), such as CLIP (J. Liu et al., 2023), BLIP (Li et al., 2022), and BiomedCLIP (Zhang et al., 2023), have demonstrated strong vision-language joint modeling capabilities through large-scale image-text pretraining. These models enable flexible and high-quality prior representations via textual encoding and interaction, offering new possibilities for medical image analysis. Unlike conventional methods, our approach leverages MLLMs to model prior knowledge on relative intensity order across different regions and intrinsic intensity characteristic around each pathology, generating robust and generalizable modality-specific intensity prior. Such textual intensity prior knowledge helps to enhance segmentation performance through tailored incorporation modules.

3. Method

3.1. Overview

Existing myocardial pathology segmentation methods largely rely on morphological and geometrical prior, which effectively captures structural information but are limited in representing intensity variations across different regions. Yet, intensity prior offers high-level semantic guidance for segmentation, which has not been fully explored. We propose the I-MMSeg method, which explicitly incorporates intensity prior into the segmentation process (see Fig. 2). First, we introduce how to generate descriptive features about intensity prior based on MLLMs (see Section 3.2). To integrate the intensity prior into the multimodal image segmentation task, we design two key modules: the intensity-prior-guided cross-modal feature enhancement module (see Section 3.3) and the intensity-prior-guided textual class feature modulation module (see Section 3.4). Both modules help to effectively improve feature representation. To the best of our knowledge, this is the first work to incorporate intensity prior into multimodal image segmentation architecture.

3.2. Modality-specific intensity prior

Traditional myocardial pathology segmentation methods often overlook modality-specific intensity variations. To address this, we use a fixed prompt template with multimodal large language models (MLLMs) (e.g., GPT-4o) to generate modality-specific textual prior that describe not only the intensity order between pathological and healthy tissues within each modality, but also the intrinsic intensity characteristic around the boundary of each pathology (see Fig. 2). To ensure accuracy and avoid semantic ambiguity, all generated textual descriptions are manually reviewed and verified by senior radiologists. Notably, such intensity prior is generated only once before training and inference, and is shared globally across all samples within each modality. Examples include, but are not limited to:

- “In the LGE modality of CMR, scar signals enhanced brightening on the myocardial, myocardium is more dark relative to the heart chamber blood signal. The scar show well-defined boundaries and sudden increase in intensity characteristics.”
- “In the T2w modality of CMR, edema signals enhanced brightening on the myocardial, myocardium is more bright relative to the heart chamber blood signal. The edema show diffuse, transitional features.”

Such textual intensity prior is encoded by the encoder Enc^T of BiomedCLIP (Zhang et al., 2023), which generates the modality-specific intensity prior embedding F^I covering all modalities and the class feature embedding F^C covering all classes:

$$F^I = \text{Enc}^T(T^I), F^C = \text{Enc}^T(T^C), \quad (1)$$

where T^I and T^C represent the textual intensity-prior description and the textual class description (e.g., This is a C class), respectively. The encoded prior F^I provides high-level semantic cues that help to enhance the model’s ability to distinguish different myocardial pathologies during segmentation.

3.3. Intensity-prior-guided cross-modal feature enhancement

To integrate intensity prior information, we introduce the **Intensity-Prior-Guided Cross-Modal Feature Enhancement (CFME)** module (see Fig. 3). This module aims to enhance the discrimination of lesion areas by using the encoded intensity prior to perform adaptive recalibration of visual feature and to facilitate deep feature interactions between different modalities. Specifically, for each modality $m \in$

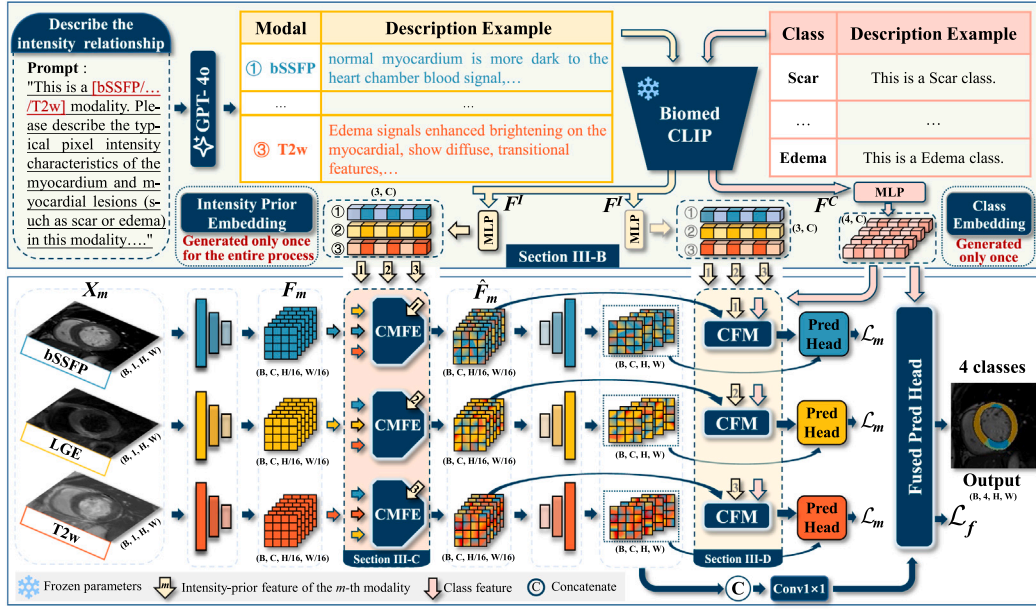


Fig. 2. Overview of the I-MMSeg method: modality-specific intensity prior generation and integration for cross-modal feature enhancement (CMFE) and textual class feature modulation (CFM). The intensity prior is generated only once at initiation and applied globally throughout the training and inference phases.

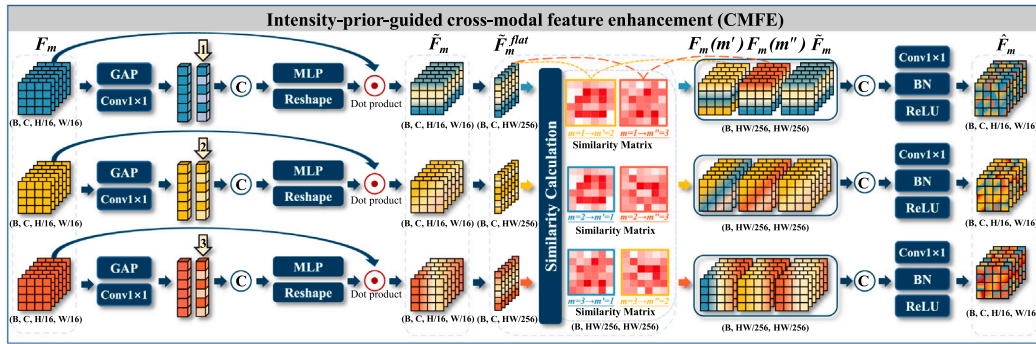


Fig. 3. Illustration of the intensity-prior-guided cross-modal feature enhancement (CMFE) module.

{bSSFP, LGE, T2w}, the input image is denoted as $X_m \in \mathbb{R}^{B \times 1 \times H \times W}$. Its visual feature F_m is then extracted using a vision encoder Enc^V :

$$F_m = \text{Enc}^V(X_m). \quad (2)$$

By integrating the visual semantic feature F_m with the corresponding encoded intensity prior F_m^I , this module dynamically generates pixel-wise weight matrix $W_m \in \mathbb{R}^{B \times C \times h \times w}$ to adaptively recalibrate the original visual features, thereby enhancing the model's ability to calibrate feature contributions based on modality-specific information. The weight matrix W_m and the recalibrated visual feature $\tilde{F}_m \in \mathbb{R}^{B \times C \times h \times w}$ are given by:

$$W_m = \sigma(\text{MLP}(\text{Concat}(\text{MLP}(F_m^I), \text{GAP}(F_m)))) \quad (3)$$

$$\tilde{F}_m = W_m \cdot F_m, \quad (4)$$

where σ represents the sigmoid activation, $\text{GAP}(\cdot)$ denotes global average pooling. B is the batch size, C is the channel, and h, w are 1/16 of the original spatial dimensions H, W .

To effectively harness the complementary nature of recalibrated multimodal features, we adopt the cross-modal attention. Specifically, the recalibrated features $\tilde{F}_m$ are reshaped into a three-dimensional matrix, as $\tilde{F}_m^{\text{flat}} \in \mathbb{R}^{B \times C \times hw}$. For any two distinct modalities $m, m' \in$

{bSSFP, LGE, T2w}, the absolute cosine similarity between their flattened feature is computed as follows:

$$\text{sim}_{m \rightarrow m'} = \left| \cos(\tilde{F}_m^{\text{flat}T}, \tilde{F}_{m'}^{\text{flat}}) \right|, \quad (5)$$

where $\tilde{F}_m^{\text{flat}T}$ is the transpose of $\tilde{F}_m^{\text{flat}}$. The $\text{sim}_{m \rightarrow m'} \in \mathbb{R}^{B \times hw \times hw}$ captures the long-range contextual correlations between different modalities at each spatial location. Guided by the computed cross-modal similarity, the recalibrated features are updated as:

$$F_m(m') = \tilde{F}_m^{\text{flat}} \cdot \text{sim}_{m \rightarrow m'}, \quad (6)$$

where the cosine similarity matrix $\text{sim}_{m \rightarrow m'}$ is first expanded along the channel dimensions to match the C channel operation, and then performs element-wise multiplication with $\tilde{F}_m^{\text{flat}}$ to achieve feature updating. The features $F_m(m')$ are reshaped to recover the original spatial dimensions. Finally, the recalibrated feature $\tilde{F}_m$ and the set of updated features are fused using a mixing function:

$$\hat{F}_m = \text{Mix}(\tilde{F}_m, F_m(m'), F_m(m'')), \quad (7)$$

where $\text{Mix}(\cdot)$ denotes a feature aggregation that concatenates the recalibrated feature $\tilde{F}_m$ and the updated features along the channel dimension, followed by a 1×1 convolution to obtain the fused feature $\hat{F}_m$. The m' and m'' denote the two modalities different from modality m . The dimensionality of $\hat{F}_m$ is consistent with the original feature F_m .

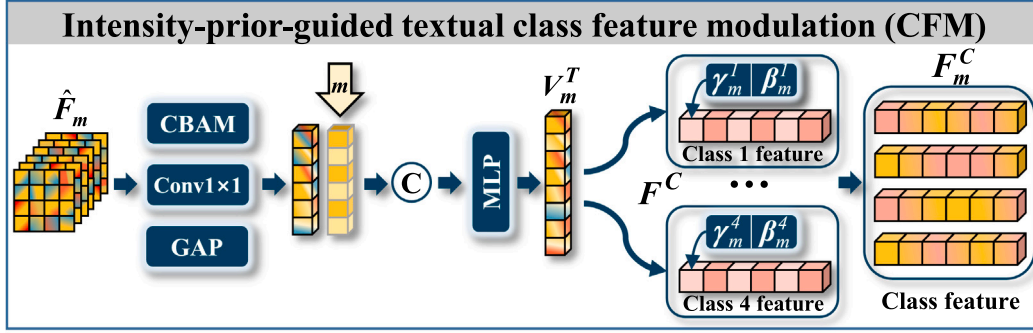


Fig. 4. Illustration of the intensity-prior-guided textual class feature modulation (CFM) module.

3.4. Intensity-prior-guided textual class feature modulation

Considering that the same semantic class exhibit different intensity characteristics within different modalities, we propose the **Intensity-Prior-Guided Textual Class Feature Modulation** module (see Fig. 4) to adaptively adjust textual class feature. Specifically, for each input modality $m \in \{\text{bSSFP, LGE, T2w}\}$, we apply the CBAM (Woo et al., 2018) module to the fused visual features $\hat{F}_m$ for feature enhancement. The resulted feature F_m^{CBAM} is fed into a 1×1 convolution for dimension reduction, followed by the global average pooling and average over the batch dimension to obtain the global visual representation $\bar{F}_m$. In each respective modality, the introduced intensity prior F_m^I is concatenated with the global visual feature $\bar{F}_m$ to form a fused representation, which is fed into a multilayer perceptron for dimensionality reduction to generate class modulation vector V_m^T for each modality. The modulation vector V_m^T is then projected to generate 8 modulation parameters, reshaped into 4 pairs of scaling and bias coefficients (γ_m^C, β_m^C) , corresponding to four class-specific textual features F^C (see Eq. (1) for background, myocardium, scar, and edema class). The modulation operation for each textual class feature F_m^C of m modality is given by:

$$F_m^C = F^C \times (1 + \gamma_m^C) + \beta_m^C. \quad (8)$$

The final modulated textual class feature F_m^C is fed into their respective prediction head. This module integrates visual feature with intensity prior to adaptively modulate textual class feature based on modality-specific intensity characteristic, helping to improve the segmentation accuracy.

3.5. Training objective

For each image X_m , we adopt the cross-entropy loss ℓ_{ce} and the Dice loss ℓ_{dc} to define the segmentation loss $\mathcal{L}_s$:

$$\mathcal{L}_s = \ell_{ce}(\hat{y}, y) + \ell_{dc}(\hat{y}, y), \quad (9)$$

where $\hat{y}$ and y is the predicted segmentation and the corresponding ground truth segmentation. This loss function is applied to both the segmentation heads for individual modalities and the fused segmentation head. The overall training objective $\mathcal{L}$ is defined by:

$$\mathcal{L} = \mathcal{L}_f + \lambda \times \sum_m \mathcal{L}_m, \quad (10)$$

where $\mathcal{L}_f$ and $\mathcal{L}_m$ denote the loss function applied to the fused segmentation head and each individual modality $m \in \{\text{bSSFP, LGE, T2w}\}$, respectively. The hyperparameter λ balances the two loss terms, and is set to 0.5 in all experiments.

4. Experiments

4.1. Datasets and evaluation protocols

We adopt a large-scale self-collected and public dataset to evaluate the proposed I-MMSeg. The details of these two datasets are given in the following.

MyoPS380 dataset. This dataset consists of multi-modality CMR images collected from 380 patients examined at Renji Hospital, Shanghai Jiao Tong University School of Medicine between 2017 and 2023. Each case contains an average of 6.42 ± 1.83 short-axis slices (per modality), covering a broad range of myocardial pathology from mild infarction to transmural myocardial injury accompanied by extensive edema.

The preprocessing workflow is illustrated in Fig. 5. To reduce variability arising from heterogeneous acquisition protocols, slice-level spatial correspondence is first established across modalities, followed by spatial normalization procedures. The original in-plane resolutions of bSSFP, LGE, and T2w measure 1.17 mm, 0.89 mm, and 0.89 mm, respectively. All images are resampled to a unified in-plane resolution of 1×1 mm. Subsequently, a myocardial-ring-mask-guided slice registration strategy was applied to enhance cross-modality alignment, where all myocardial ring masks were obtained using a pre-trained SAMed (Zhang and Liu, 2023). The final volumes are center-cropped to 128×128 to ensure spatial consistency and focus on the myocardial region. Myocardial pathology annotations are independently performed on modality-specific images by three experienced radiologists and subsequently reviewed by a senior radiologist to ensure reliability and consistency. The dataset is divided into 304 training cases and 76 testing cases. To the best of our knowledge, the MyoPS380 dataset represents the largest publicly curated multi-modality CMR dataset for myocardial pathology segmentation. It serves as a comprehensive resource for developing and evaluating algorithms designed for multi-modality fusion and myocardial tissue characterization. Additional details are available at: https://github.com/zzzzzzl24/1_MMSeg.

MyoPS++ dataset (Ding et al., 2023). The MyoPS++ dataset, provided by the CARE 2024 MyoPS++ Challenge, comprises data from 250 patients collected across seven medical centers in China, France, and the United Kingdom. Among the data for these patients, only 95 patients with complete bSSFP, LGE, and T2w modalities are publicly available. Therefore, we select the data for these 95 patients for validation. We randomly split this subset into 76 cases for training and 19 cases for testing.

Evaluation Protocols. The proposed method is evaluated with two widely used metrics in myocardial pathology segmentation task: Dice coefficient (Dice) and the Hausdorff Distance (HD).

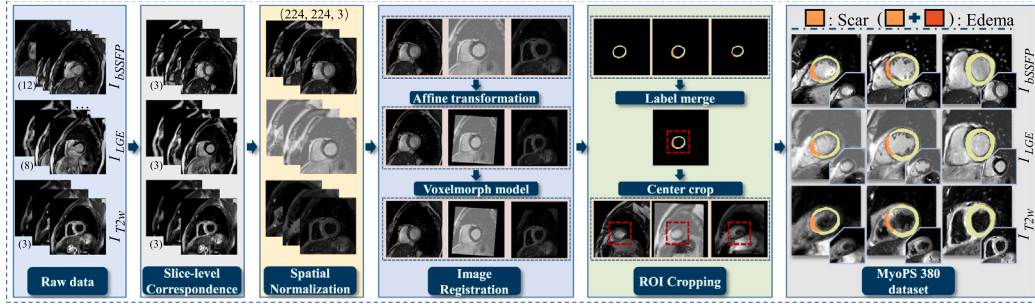


Fig. 5. Preprocessing pipeline for the self-collected MyoPS380 dataset.

Table 1

Quantitative comparison between I-MMSeg and state-of-the-art segmentation methods on the self-collected MyoPS380 dataset.

Methods		Scar		Edema		Myocardium		Avg. pathology	
		Dice↑	HD↓	Dice↑	HD↓	Dice↑	HD↓	Dice↑	HD↓
Mono-modal	LGE-TransUNet (Chen et al., 2024)	61.00	5.11	62.23	6.61	78.26	1.61	61.62	5.86
	T2w-TransUNet (Chen et al., 2024)	45.94	10.39	63.67	7.30	77.76	1.61	54.81	8.85
Early-fusion	U-Net (Ronneberger et al., 2015)	68.92	6.61	71.05	5.41	85.14	1.21	69.99	6.01
	nnU-Net (Isensee et al., 2021)	71.21	4.68	73.07	5.12	86.23	1.11	72.14	4.90
	TransUNet (Chen et al., 2024)	63.90	6.45	71.12	5.46	83.67	1.20	67.51	5.96
	SAMed (Zhang and Liu, 2023)	68.99	3.68	72.58	5.13	83.20	1.23	70.79	4.41
	MyoPS_Challenge_1 (Zhai et al., 2020)	71.15	5.08	72.63	4.39	85.74	1.18	71.89	4.74
	MyoPS_Challenge_4 (Ankenbrand et al., 2020)	71.60	4.98	72.55	5.23	86.44	1.11	72.08	5.11
Mid-level feature-fusion	mmFormer (Zhang et al., 2022)	70.18	5.12	71.96	5.76	84.66	1.32	71.07	5.44
	M ² FTrans (Shi et al., 2023)	71.45	3.86	72.96	4.60	85.35	1.16	72.21	4.23
	IMS ² Trans (Zhang et al., 2024)	69.84	6.24	72.05	5.46	84.34	1.44	70.95	5.85
	Stacked BCDU-Net (Martín-Isla et al., 2020)	70.43	4.91	71.68	5.82	84.64	1.32	71.56	5.37
	EfficientSeg (J. Zhang et al., 2020)	70.04	5.28	73.15	5.40	86.55	1.10	71.60	5.34
	MyoPS_Challenge_5 (Z. Zhang et al., 2020)	71.24	4.52	72.12	5.18	85.30	1.16	71.68	4.85
	MyoPS-Net (Qiu et al., 2023)	69.55	5.63	72.14	5.39	83.97	1.20	70.85	5.51
	AWSnet (Wang et al., 2022)	69.89	5.21	71.94	5.75	84.32	1.18	70.92	5.48
	I-MMSeg (Ours)	73.64	3.66	75.44	3.89	87.41	1.07	74.54	3.78

4.2. Implementation details

All experiments are conducted on a server with two NVIDIA RTX 4090 GPUs (24 GB each). We employ ResNet-50 (He et al., 2016) model as the backbone of our segmentation network. The proposed method is trained using the AdamW optimizer with an initial learning rate of 0.001, a weight decay of 10^{-4} , and a polynomial learning rate decay strategy. We train the model on 128×128 images with standard normalization and common data augmentation techniques, including random rotation, flipping, Gaussian noise, and random shifts. The batch size is set to 48. We train the segmentation model for 300 epochs.

4.3. Comparative evaluation of the proposed I-MMSeg

Comparative methods: To illustrate the importance of multi-modalities for myocardial pathology segmentation and to validate the effectiveness of I-MMSeg, we conducted a comprehensive comparison on the MyoPS380 and MyoPS++ datasets against three types of approaches: single-modality models, early-fusion models, and mid-level feature-fusion models. Among the early fusion methods, U-Net (Ronneberger et al., 2015) and nnU-Net (Isensee et al., 2021) serve as classical architectures for medical image segmentation and consistently exhibit strong performance. TransUNet (Chen et al., 2024) integrates Transformer modules into the U-Net architecture, effectively addressing U-Net's limitation in modeling long-range dependencies. SAMed (Zhang and Liu, 2023), as a general-purpose solution for medical image segmentation, demonstrates strong performance in tasks involving multi-organ segmentation. For feature-level multimodal image segmentation methods, mmFormer (Zhang et al., 2022), M²FTrans (Shi et al., 2023), and IMS²Trans (Zhang et al., 2024) leverage advanced Transformer architectures to build both local and global contextual representations

within each modality while capturing long-range dependencies across modalities. These methods are widely applied in multimodal image segmentation tasks.

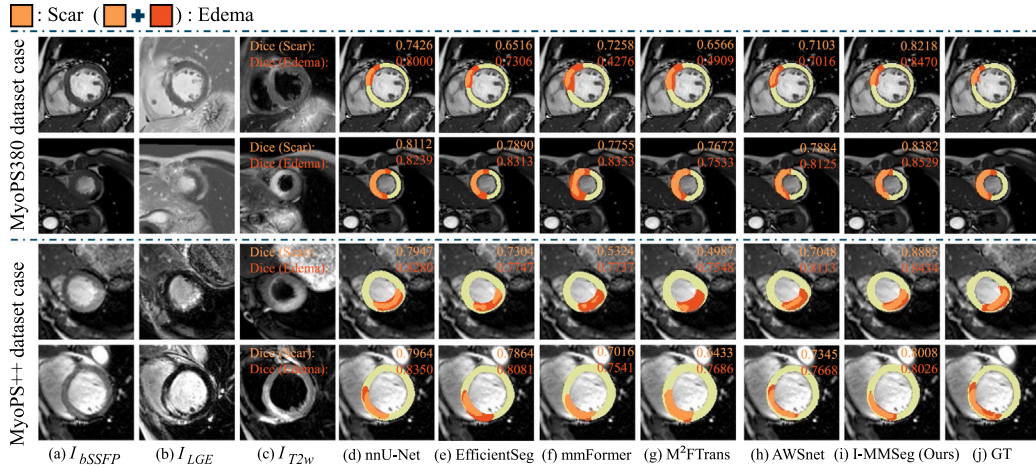
We also incorporated five top-performing methods from the MyoPS Challenge (Li et al., 2023b). Methods without official names are collectively denoted as MyoPS_Challenge_{i}, i denotes the actual ranking. Among them, EfficientSeg (J. Zhang et al., 2020) serves as a representative solution from the challenge. Its decoder employs a weighted bidirectional feature pyramid network, which enables efficient multi-scale feature fusion. In addition, MyoPS-Net (Qiu et al., 2023) and AWSnet (Wang et al., 2022) represent two frameworks specifically developed for the myocardial pathology segmentation task in recent years. MyoPS-Net (Qiu et al., 2023) adopts an end-to-end deep neural architecture capable of handling varying numbers of CMR images and complex modality combinations. AWSnet (Wang et al., 2022) employs a coarse-to-fine two-stage segmentation strategy to enhance segmentation accuracy for pathological regions.

Comparative results: The quantitative evaluations on MyoPS380 and MyoPS++ are depicted in Tables 1 and 2, respectively. The results clearly demonstrate that multimodal fusion methods significantly outperform mono-modal segmentation on both datasets, confirming the necessity of integrating multimodal information. Classical architectures such as U-Net (Ronneberger et al., 2015) and nnU-Net (Isensee et al., 2021) still exhibit stable and reliable performance across both datasets. In particular, nnU-Net consistently maintains high Dice scores across different tissue types, reflecting its strong generalization capability. In addition, the five top-performing methods from the first MyoPS challenge also achieve favorable results, among which EfficientSeg (J. Zhang et al., 2020) demonstrates strong adaptability and solid performance on both datasets. Our proposed I-MMSeg method achieves the highest Dice score and the lowest HD across all pathological categories,

Table 2

Quantitative comparison between I-MMSeg and state-of-the-art segmentation methods on the MyoPS++ dataset.

Methods		Scar		Edema		Myocardium		Avg. pathology	
		Dice↑	HD↓	Dice↑	HD↓	Dice↑	HD↓	Dice↑	HD↓
Mono-modal	LGE-TransUNet (Chen et al., 2024)	42.01	12.16	36.21	10.82	76.92	2.20	39.11	11.49
	T2w-TransUNet (Chen et al., 2024)	36.25	13.59	62.16	9.79	82.78	1.64	49.21	11.69
Early-fusion	U-Net (Ronneberger et al., 2015)	57.85	8.96	67.31	10.24	83.02	2.54	62.58	9.60
	nnU-Net (Isensee et al., 2021)	70.89	3.79	72.93	4.61	89.89	1.07	71.91	4.20
	TransUNet (Chen et al., 2024)	59.40	8.40	68.79	7.63	86.69	1.29	64.10	8.02
	SAMed (Zhang and Liu, 2023)	56.32	7.91	68.79	7.54	83.20	1.63	62.56	7.73
	MyoPS_Challenge_1 (Zhai et al., 2020)	68.46	3.36	71.30	5.06	89.18	1.14	69.88	4.21
	MyoPS_Challenge_4 (Ankenbrand et al., 2020)	66.44	4.30	69.87	6.90	88.73	<u>1.10</u>	68.16	5.60
Mid-level feature-fusion	mmFormer (Zhang et al., 2022)	71.24	3.32	73.74	5.02	88.47	1.07	72.49	4.17
	M ² FTrans (Shi et al., 2023)	69.97	3.57	72.82	5.21	87.30	1.13	71.40	4.39
	IMS ² Trans (Zhang et al., 2024)	65.33	5.56	70.58	7.29	87.37	1.24	67.96	6.43
	Stacked BCDU-Net (Martín-Isla et al., 2020)	68.76	6.52	72.32	6.92	86.62	1.32	70.54	6.72
	EfficientSeg (J. Zhang et al., 2020)	70.57	3.37	73.70	6.84	89.04	1.07	72.14	5.11
	MyoPS_Challenge_5 (Z. Zhang et al., 2020)	57.42	8.52	64.05	11.35	81.62	3.14	60.74	9.94
	MyoPS-Net (Qiu et al., 2023)	69.67	5.19	72.22	5.55	86.63	1.28	70.95	5.37
	AWSnet (Wang et al., 2022)	69.75	3.67	72.33	5.32	87.51	1.12	71.04	4.50
	I-MMSeg (Ours)	74.33	3.04	75.90	3.85	<u>89.73</u>	<u>1.10</u>	75.12	3.45

**Fig. 6.** Some qualitative segmentation results of myocardial pathology on the self-collected MyoPS380 dataset (top two rows) and MyoPS++ dataset (bottom two rows).

indicating that it produces more accurate and smoother segmentation results. Specifically, on the MyoPS380 dataset, I-MMSeg obtains an average pathology Dice score of 74.54%, outperforming the second-best method M²FTrans (72.21%) by 2.33%, and reduces the average HD from 4.23 to 3.78, the lowest among all competitors. Similarly, on the MyoPS++ dataset, I-MMSeg achieves an average Dice of 75.12%, surpassing the second-best mmFormer (72.49%) by 2.63%, and achieves the best HD (3.45) across all pathological classes. Notably, I-MMSeg achieves the best Dice score and HD for all two pathological categories (scar, edema) in both datasets, highlighting its robustness and generalizability. For example, on MyoPS++, it achieves 74.33% Dice for scar (vs. 71.24% by mmFormer), and 75.90% for edema (vs. 73.74%), while consistently reducing HD.

This performance gain is mainly attributed to the proposed incorporation of intensity prior, which enables the network to effectively capture modality-specific contrast patterns and enhances feature discrimination across modalities. Qualitative results of the top six models in overall lesion-segmentation performance on the two datasets (see Fig. 6) further demonstrate the superior accuracy and smoother contours achieved by our I-MMSeg.

4.4. Ablation study

We conduct comprehensive ablation studies to assess the individual contribution of the proposed intensity prior, CMFE module, and CFM

module to the overall model performance. The architectural configurations of different alternatives are illustrated in Fig. 7. The Baseline model comprises three independent encoder-decoder branches followed by fused prediction head. The model 2 and Model 4 are obtained by adding the CMFE and CFM modules to the Baseline model, respectively. Model 6 incorporates both the CMFE and CFM modules to the Baseline model. In addition, Model 3, 5, and 7 are built upon Model 2, 4, and 6, respectively, by introducing the intensity prior, allowing us to verify the effectiveness of intensity prior in improving myocardial pathology segmentation accuracy.

The corresponding results are summarized in Table 3. On the MyoPS++ dataset, introducing the CMFE and CFM module alone increases the average Dice score for pathological regions from the Baseline of 71.77% to 72.69% and 72.77%, respectively. When combined together, these modules further improve the performance to 73.92%, indicating their complementary strengths in feature enhancement and semantic modeling. When incorporating the intensity prior, the final model I-MMSeg (Model 7) achieves a significantly higher average Dice score of 75.12%, obtaining a 3.35% Dice improvement over the Baseline. This highlights the critical role of the intensity prior incorporation in enhancing feature discriminability and leveraging cross-modal synergies. A similar trend is observed on the MyoPS380 dataset. It is noteworthy that the improvement on the smaller dataset MyoPS++ is more significant (3.35% VS. 1.85%) than that on the larger MyoPS380 dataset. This further demonstrates the importance of integrating the intensity

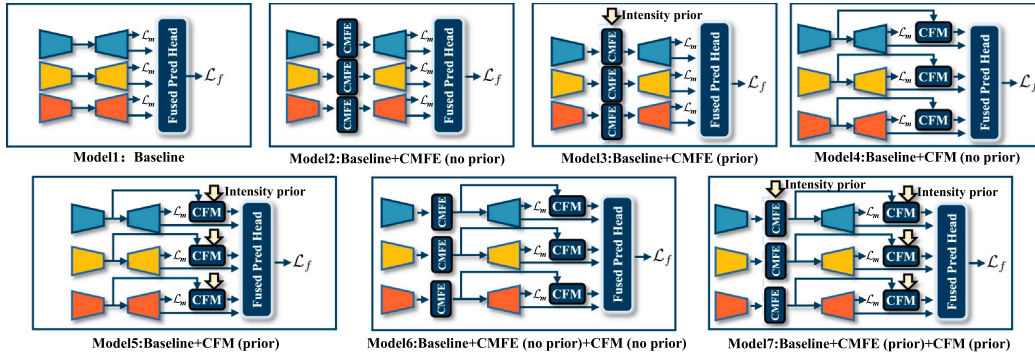


Fig. 7. Configuration overview of different ablation models with/without CMFE, CFM, and intensity prior.

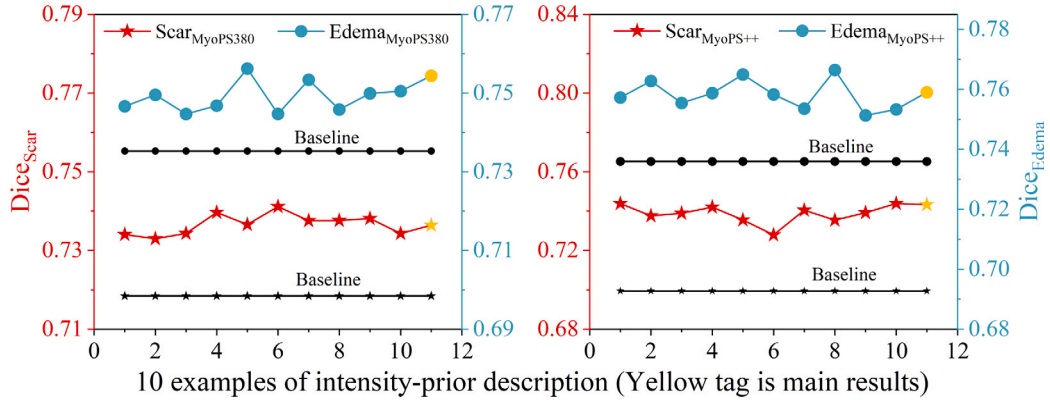


Fig. 8. Quantitative evaluation based on Dice score on using ten different intensity prior descriptions across two datasets.

Table 3

Ablation study results on the MyoPS380 and MyoPS++ datasets. We evaluate the contribution of the proposed CMFE module and CFM module. We also evaluate the contribution of intensity prior guidance.

Methods		Scar		Edema		Myocardium		Avg. pathology	
		Dice↑	HD↓	Dice↑	HD↓	Dice↑	HD↓	Dice↑	HD↓
MyoPS380 dataset	Baseline	71.84	4.43	73.53	4.38	86.94	1.10	72.69	4.41
	+CMFE (no prior)	72.43	3.85	74.65	3.96	87.03	1.08	73.54	3.91
	+CMFE	72.96	3.71	74.85	3.88	87.25	1.08	73.91	3.80
	+CFM (no prior)	72.54	3.83	73.93	4.11	86.96	1.10	73.24	3.97
	+CFM	72.96	3.72	74.91	3.85	87.22	1.10	73.94	3.79
	+CMFE (no prior) +CFM (no prior)	72.43	3.99	74.54	3.97	87.14	1.08	73.49	3.98
	+CMFE +CFM	73.64	3.66	75.44	3.89	87.41	1.07	74.54	3.78
MyoPS++ dataset	Baseline	69.94	3.22	73.59	4.90	89.64	1.07	71.77	4.06
	+CMFE (no prior)	70.81	4.03	74.56	4.24	89.68	1.04	72.69	4.14
	+CMFE	72.77	3.14	74.69	4.42	89.63	1.10	73.73	3.78
	+CFM (no prior)	71.89	3.92	73.65	4.44	89.52	1.09	72.77	4.18
	+CFM	72.39	3.31	74.49	4.14	89.65	1.10	73.44	3.73
	+CMFE (no prior) +CFM (no prior)	72.82	3.46	75.01	4.43	89.92	1.10	73.92	3.95
	+CMFE +CFM	74.33	3.04	75.90	3.85	89.73	1.10	75.12	3.45

prior, especially on small training datasets, where prior knowledge helps more.

Furthermore, we also investigate the adaptability of I-MMSeg to varying intensity prior. Based on the original intensity prior descriptions, we used GPT-4o to generate ten alternative sets of structurally different but semantically similar descriptions. As illustrated in Fig. 8, despite structurally different descriptions, I-MMSeg maintains similarly superior performance compared to the Baseline model, highlighting its strong adaptability and robustness to slight changes in intensity prior description.

4.5. Assessment under missing modality scenarios

Missing modalities are frequently encountered in clinical practice, often leading to significant degradation in segmentation performance. To comprehensively evaluate the robustness of the proposed I-MMSeg method under various modality combinations, we adopt the modality masking strategy from RFNet (Ding et al., 2021) and design seven different modality-missing configurations for testing. Specifically, during training, a random missing-modality configuration is selected at

Table 4

Quantitative comparison between our I-MMSeg and SOTA methods trained on MyoPS380 dataset with randomly missing modalities. During inference, available and missing modalities are denoted by • and ○, respectively.

Modality			Dice↑				Scar				HD↓				Edema				HD↓			
<i>bSSFP</i>	<i>LGE</i>	<i>T2w</i>	mmF	M ² FT	IMS ² T	Ours	mmF	M ² FT	IMS ² T	Ours	mmF	M ² FT	IMS ² T	Ours	mmF	M ² FT	IMS ² T	Ours	mmF	M ² FT	IMS ² T	Ours
•	○	○	<u>44.13</u>	42.01	43.12	50.74	<u>11.30</u>	12.84	14.79	9.90	<u>50.72</u>	49.58	49.46	59.96	8.91	<u>8.83</u>	10.27	8.16				
○	•	○	65.49	65.42	63.45	70.36	5.72	6.24	7.53	4.02	61.70	60.91	60.35	66.48	7.26	8.06	7.54	5.43				
○	○	•	49.21	<u>49.45</u>	47.19	53.59	<u>11.06</u>	11.39	11.71	9.02	64.91	<u>65.24</u>	63.22	68.95	<u>7.55</u>	8.21	<u>7.55</u>	6.39				
•	•	○	70.67	<u>70.77</u>	70.09	72.02	4.84	4.80	5.34	3.61	66.58	65.89	65.97	69.60	5.98	6.65	<u>5.46</u>	4.82				
•	○	•	<u>54.70</u>	54.36	53.64	57.48	<u>9.31</u>	10.10	10.36	7.44	<u>68.98</u>	68.09	67.36	71.87	<u>5.64</u>	6.19	6.61	5.59				
○	•	•	67.85	67.82	<u>68.00</u>	71.48	<u>4.44</u>	5.51	5.03	4.05	<u>70.12</u>	69.74	68.91	73.50	<u>5.97</u>	6.78	7.27	4.13				
•	•	•	<u>71.03</u>	70.86	70.97	72.20	3.77	4.79	4.34	4.14	<u>72.04</u>	71.34	71.05	74.63	<u>4.96</u>	6.19	5.71	4.00				
Average			<u>60.44</u>	60.10	59.49	63.98	<u>7.21</u>	7.95	8.44	6.03	<u>65.01</u>	64.40	63.76	69.28	<u>6.61</u>	7.27	7.20	5.50				

Table 5

Quantitative comparison between our I-MMSeg and SOTA methods trained on MyoPS++ dataset with randomly missing modalities. During inference, available and missing modalities are denoted by • and ○, respectively.

Modality			Dice↑				Scar				HD↓				Edema				HD↓			
<i>bSSFP</i>	<i>LGE</i>	<i>T2w</i>	mmF	M ² FT	IMS ² T	Ours	mmF	M ² FT	IMS ² T	Ours	mmF	M ² FT	IMS ² T	Ours	mmF	M ² FT	IMS ² T	Ours	mmF	M ² FT	IMS ² T	Ours
•	○	○	<u>46.12</u>	45.04	43.16	52.05	<u>12.48</u>	13.42	14.85	9.97	<u>53.86</u>	53.62	52.26	60.39	<u>13.48</u>	13.65	14.11	8.49				
○	•	○	66.00	<u>67.29</u>	65.54	71.49	6.04	4.94	6.21	3.83	63.75	<u>63.76</u>	61.11	64.80	8.85	<u>8.80</u>	9.25	7.60				
○	○	•	45.33	48.86	48.22	43.43	13.37	<u>11.39</u>	11.54	9.13	63.92	<u>64.75</u>	62.80	67.74	10.00	<u>9.55</u>	10.62	6.89				
•	•	○	<u>71.35</u>	70.03	70.32	72.49	3.66	4.44	4.02	<u>3.68</u>	<u>65.04</u>	64.46	63.60	68.12	<u>8.53</u>	9.22	9.23	6.24				
•	○	•	51.59	53.77	53.24	51.17	11.68	8.58	9.82	6.21	<u>66.62</u>	66.26	66.50	68.70	8.57	<u>7.30</u>	8.57	6.54				
○	•	•	69.09	<u>69.52</u>	68.81	71.43	<u>3.82</u>	4.65	4.36	3.48	<u>72.60</u>	72.36	70.99	75.15	6.19	<u>5.02</u>	6.99	4.63				
•	•	•	71.86	70.65	70.37	<u>71.59</u>	3.31	4.26	4.55	<u>3.52</u>	<u>72.59</u>	71.42	71.85	74.90	5.82	<u>5.46</u>	5.51	5.15				
Average			60.19	<u>60.74</u>	59.95	61.95	7.77	<u>7.38</u>	7.91	5.69	<u>65.48</u>	65.24	64.16	68.55	8.78	<u>8.43</u>	9.18	6.51				

Table 6

Robustness comparison between I-MMSeg and state-of-the-art methods under varying LGE absence levels during training and testing.

Absence ratio	Methods	Scar		Edema		Myocardium		Avg. pathology	
		Dice↑	HD↓	Dice↑	HD↓	Dice↑	HD↓	Dice↑	HD↓
10%	mmFormer (Zhang et al., 2022)	69.10	4.85	72.29	5.35	84.70	1.30	70.70	5.10
	M ² FTrans (Shi et al., 2023)	69.68	4.23	72.74	4.97	85.00	1.18	<u>71.21</u>	<u>4.60</u>
	IMS ² Trans (Zhang et al., 2024)	68.42	6.30	72.54	5.04	84.36	1.24	70.48	5.67
	I-MMSeg (Ours)	72.37	3.91	74.77	3.87	87.07	1.09	73.57	3.89
40%	mmFormer (Zhang et al., 2022)	65.19	5.89	72.28	5.31	84.73	1.61	68.74	5.60
	M ² FTrans (Shi et al., 2023)	65.65	5.40	72.05	5.17	84.95	1.17	<u>68.85</u>	<u>5.29</u>
	IMS ² Trans (Zhang et al., 2024)	64.22	6.96	70.80	5.49	84.30	1.54	67.51	6.23
	I-MMSeg (Ours)	67.66	4.52	73.51	3.77	86.69	1.09	70.59	4.15
70%	mmFormer (Zhang et al., 2022)	59.95	8.12	71.28	5.06	84.39	1.20	<u>65.62</u>	6.59
	M ² FTrans (Shi et al., 2023)	57.44	7.60	70.96	4.96	84.60	1.17	<u>64.20</u>	<u>6.28</u>
	IMS ² Trans (Zhang et al., 2024)	58.78	8.17	70.10	5.50	84.26	1.22	64.44	6.84
	I-MMSeg (Ours)	63.52	6.16	72.97	4.41	86.43	1.11	68.25	5.29
100%	mmFormer (Zhang et al., 2022)	52.65	10.56	68.74	6.09	84.33	1.23	60.70	8.33
	M ² FTrans (Shi et al., 2023)	54.55	9.59	69.12	5.79	84.39	1.19	<u>61.84</u>	<u>7.69</u>
	IMS ² Trans (Zhang et al., 2024)	52.33	9.51	67.54	5.95	84.00	1.27	59.94	7.73
	I-MMSeg (Ours)	58.27	7.00	71.63	5.17	86.31	1.12	64.95	6.09

each data loading step to simulate the diverse modality absence scenarios that may occur in real-world settings. Using the MyoPS380 and MyoPS++ datasets, we conducted a systematic comparison between I-MMSeg and three representative approaches for handling missing modalities: mmFormer (Zhang et al., 2022), M²FTrans (Shi et al., 2023), and IMS²Trans (Zhang et al., 2024).

As presented in Tables 4 and 5, the results show that I-MMSeg consistently achieves best or second-best scar segmentation performance in most configurations and outperforms existing methods in both Dice and HD metrics for edema segmentation. Notably, under severe modality absence (e.g., only bSSFP available), I-MMSeg demonstrates notable robustness in both scar and edema segmentation tasks, achieving Dice score of 50.74% and 59.96% on the MyoPS380 dataset, and 52.05% and 60.39% on the MyoPS++ dataset, respectively, all significantly outperforming existing methods. Across all seven modality-missing configurations, I-MMSeg achieves the highest average Dice score of 63.98%/69.28% (scar/edema) on MyoPS380

and 61.95%/68.55% (scar/edema) on MyoPS++, confirming the effectiveness of the proposed method under multiple missing modality configurations.

We further simulated an LGE-missing scenario that better reflects real-world clinical practice using the self-collected MyoPS380 dataset. Specifically, only the LGE modality was absent during both training and testing, with absence ratios of 10%, 40%, 70%, and 100%. The quantitative results under different levels of LGE absence are reported in Table 6. Under all LGE-missing conditions, I-MMSeg achieves the best segmentation performance. Notably, its advantage becomes more pronounced at higher missing rates (70%–100%). For instance, compared with the second-best method, the improvement in average pathology segmentation increases from 2.36% at a 10% LGE absence rate to 3.11% at a 100% LGE absence rate. These results demonstrate the robustness of I-MMSeg under clinically realistic LGE-missing scenarios.

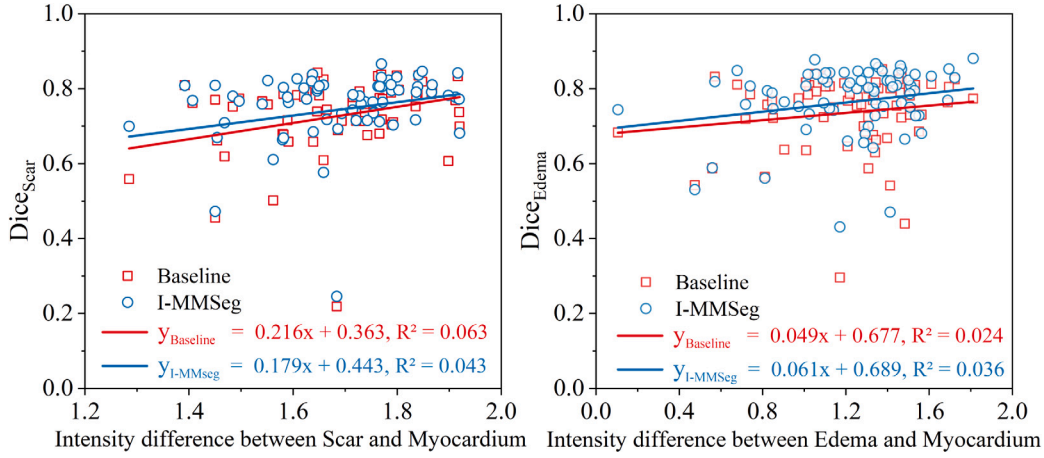


Fig. 9. Correlation analysis between segmentation accuracy and intensity differences of lesion and myocardium in each sample. The horizontal coordinate corresponds to the logarithm (base 10) of the pixel intensity differences.

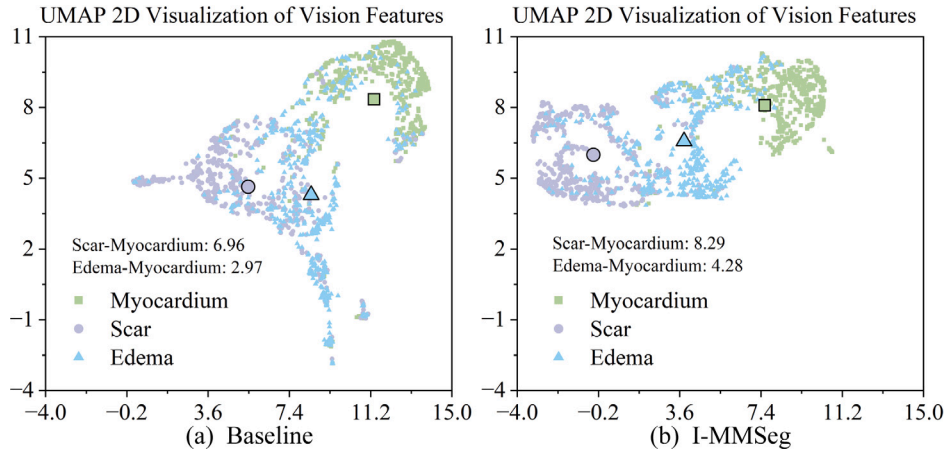


Fig. 10. Comparison on UMAP visualization of vision features from the baseline model and our I-MMSeg, characterizing the low-dimensional feature input to the fused head.

5. Discussion

Extensive experiments have validated the effectiveness of the proposed method in multimodal myocardial pathology segmentation tasks. To further investigate the contribution of incorporating intensity prior, we analyze the relationship between pixel intensity differences of lesion and healthy myocardium regions and segmentation accuracy. As shown in Fig. 9, for the scar segmentation, when the intensity contrast is low, I-MMSeg demonstrates a more significant improvement in segmentation performance. This is mainly because that scars often have relatively clear boundaries and local intensity discontinuities. Under low contrast conditions, the model struggles to achieve precise recognition based solely on visual feature. The intensity prior effectively compensates for the missing boundary cues, guiding the model to focus better on potential lesion areas, thereby enhancing scar discrimination. In contrast, the edema exhibits diffuse boundaries and irregular morphology, with gradual intensity transitions and a lack of clear structural edges. Such ambiguous features make the model prone to confusion within high-response regions lacking distinct boundaries, leading to misclassification. The intensity prior provides semantic cues such as “blurred boundaries” and “diffuse enhancement”, enabling the model to selectively extract lesion-consistent response patterns. This semantic constraint becomes more significant when pixel differences and interference signals are stronger in edema regions, further improving segmentation performance.

Although the mechanisms by which the intensity prior influences scar and edema regions differ, the notable performance gains indicate that incorporating intensity prior effectively enhances the semantic separability between lesion and healthy myocardium regions. To further verify this, we perform UMAP (McInnes et al., 2018) visualization (with parameters $n_neighbors = 30$, $min_dist = 0.01$) on the input feature of the fused segmentation head from both the baseline and our I-MMSeg model. As shown in Fig. 10, the results clearly demonstrate the feature distribution of three classes. Compared to the baseline model, features extracted by I-MMSeg exhibit clearer inter-class separation, reflecting the model’s improved ability to distinguish subtle differences among various lesion types and healthy tissue. With the introduction of intensity prior, the model forms more compact clusters in the high-dimensional feature space, reducing class overlap, alleviating ambiguity caused by fuzzy boundaries, and significantly enhancing segmentation accuracy and stability.

A potential limitation of the proposed method is that it requires more computation time compared to the baseline model, due to the computation of extra textual feature on intensity prior and two introduced modules incorporating such intensity prior. Yet, fortunately, the textual feature on intensity prior with multimodal large language models (e.g., GPT-4o) only needs to be computed once for both training and testing, and can be computed offline. Therefore, this does not influence the runtime in practice. The introduced CFME and CMF modules are lightweight compared to the backbone network. The proposed I-MMSeg

increases the average runtime from 0.267 s for the baseline model to 0.294 s for a $128 \times 128 \times 6$ volume.

6. Conclusion

This study presents a novel intensity-prior-guided multimodal image segmentation method for myocardial pathology segmentation. To the best of our knowledge, this is the first work to systematically incorporate modality-specific intensity prior into myocardial pathology segmentation. We leverage a multimodal large language model to generate interpretable textual descriptions on the relative intensity orders across different regions and the intrinsic intensity characteristic around the boundary of each pathology, serving as high-level prior knowledge. The proposed I-MMSeg introduces two key modules to incorporate such prior knowledge: intensity-prior-guided cross-modal feature enhancement and textual class feature modulation module. These two modules jointly facilitate effective multimodal information integration and intensity prior incorporation, enhancing the identification of pathological regions. Comprehensive experiments validate the superior performance of our I-MMSeg on myocardial pathology segmentation, significantly outperforming existing methods. In the future, we would like to explore the proposed intensity-prior-guided segmentation method to broader multimodal medical image segmentation tasks, while also further exploring more robust and richer automatically derived prior descriptions.

CRedit authorship contribution statement

Donggen Fang: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yuliang Gu:** Visualization, Validation, Software, Methodology, Formal analysis. **Lingyi Yu:** Visualization, Validation, Formal analysis, Data curation. **Bo Du:** Writing – review & editing, Methodology, Formal analysis. **Yongchao Xu:** Writing – review & editing, Visualization, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Lianming Wu:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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